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Exercise 10.4

1) Evaluate the following:

a) $\int x^2 \ln x \, dx$

$= \int \ln x \cdot x^2 \, dx$

$= \ln x \int x^2 dx - \int \left\{ \frac{d \ln x}{dx} \cdot \int x^2 dx \right\} dx$

$= \ln x \times \frac{x^3}{3} - \int \left(\frac{1}{x} \times \frac{x^3}{3} \right) dx$

$= \ln x \times \frac{x^3}{3} - \frac{1}{3} \frac{x^3}{3} + c$

$= \frac{x^3}{3} \left(\ln x - \frac{1}{3} \right) + c$

b) $\int \ln x \, dx$

$= \int 1 \cdot \ln x \, dx$

$= \int \ln x \cdot 1 \, dx$

$= \ln x \int 1 \, dx - \int \left\{ \frac{d \ln x}{dx} \times \int 1 \, dx \right\} dx$

$= \ln x \times x - \int \left\{ \frac{1}{\cancel{x}} \times \cancel{x} \, dx \right\}$

$= x \ln x - x + c$

c) $\int (\ln x)^2 \, dx$

$= \int (\ln x)^2 \cdot 1 \, dx$

$= (\ln x)^2 \int 1 \, dx - \int \left\{ \frac{d (\ln x)^2}{dx} \times \int 1 \, dx \right\} dx$

$= (\ln x)^2 \times x - \int \left\{ \left(\frac{d (\ln x)^2}{d \ln x} \times \frac{d \ln x}{dx} \right) \times x \right\} dx$

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$= x(\ln x)^2 - \int 2\ln x \times \frac{1}{\cancel{x}} \times \cancel{x}\, dx$

$= x(\ln x)^2 - 2\int \ln x\, dx$

$= x(\ln x)^2 - 2\{x\ln x - x + c\}$

$= x(\ln x)^2 - 2x\ln x + 2x + c$

2) Evaluate the following

a) $\int x e^{5x}\, dx$

$= x\int e^{5x} dx - \int\left\{\frac{dx}{dx}\int e^{5x} dx\right\} dx$

$= x \times \frac{e^{5x}}{5} - \int 1 \times \frac{e^{5x}}{5}\, dx$

$= \frac{xe^{5x}}{5} - \frac{1}{5}\int\left[e^{5x} dx\right]$

$= \frac{xe^{5x}}{5} - \frac{1}{5} \times \frac{e^{5x}}{5}$

$= \frac{e^{5x}}{5}\left[\frac{5x-1}{5}\right] + c$

b) $\int (2x-1)e^{2x}\, dx$

$= (2x-1)\int e^{2x} dx - \int\left\{\frac{d(2x-1)}{dx} \times \int e^{2x} dx\right\} dx$

$= (2x-1) \times \frac{e^{2x}}{2} - \int \cancel{2} \times \frac{e^{2x}}{\cancel{2}}\, dx$

$= (2x-1) \times \frac{e^{2x}}{2} - \frac{e^{2x}}{2} + c$

$= \frac{e^{2x}}{2}[2x-1+1] + c$

$= \frac{e^{2x}}{\cancel{2}} \times \cancel{2}(x-1) + c$

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$= \cancel{x}(x-1)e^{2x} + c$

c) $\int x^2 e^x \, dx$

$= x^2 \int e^x dx - \int \left\{ \frac{dx^2}{dx} \times \int e^x dx \right\} dx$

$= x^2 \times e^x - \int 2x \times e^x \, dx$

$= x^2 . e^x - 2 \left[x \int e^x dx - \int \left\{ \frac{dx}{dx} \times \int e^x dx \right\} dx \right]$

$= x^2 . e^x - 2 \left[x . e^x - \int 1 \times e^x \, dx \right]$

$= x^2 . e^x - 2 \left(x e^x - e^x \right) + c$

$= x^2 . e^x - 2x e^x + 2e^x + c$

③ Evaluate the following

a) $\int x \cos x \, dx$

$= x \int \cos x \, dx - \int \left\{ \frac{dx}{dx} \cdot \int \cos x \, dx \right\} dx$

$= x(+\sin x) - \int 1 \cdot (+\sin x) \, dx$

$= x \sin x - \int \sin x \, dx$

$= x \sin x - (-\cos x) + c$

$= x \sin x + \cos x + c$

b) $\int x \sin ax \, dx$

$= x \int \sin ax \, dx - \int \left\{ \frac{dx}{dx} \int \sin ax \, dx \right\} dx$

$= x \left(-\frac{\cos ax}{a} \right) - \int 1 \times \left(-\frac{\cos ax}{a} \right) dx$

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$= \frac{-x}{a}\cos ax + \frac{1}{a}\int \cos ax\, dx$

$= \frac{-x}{a}\cos ax + \frac{1}{a}\frac{\sin ax}{a} + c$

$= -\frac{x}{a}\cos ax + \frac{1}{a^2}\sin ax + c$

$= \frac{1}{a^2}\sin ax - \frac{x}{a}\cos ax + c$

c) $\int x \sec^2 ax\, dx$

$= x\int \sec^2 ax\, dx - \int\left\{\frac{dx}{dx}\int \sec^2 ax\, dx\right\} dx$

$= x\frac{\tan ax}{a} - \int 1 \cdot \frac{\tan ax}{a}\, dx$

$= \frac{x}{a}\tan ax - \frac{1}{a}\frac{\ln(\sec ax)}{a} + c$

$= \frac{x}{a}\tan ax - \frac{1}{a^2}\ln|\sec ax| + c$

d) $\int x \operatorname{cosec} x \cot x\, dx$

$= x\int \operatorname{cosec} x . \cot x\, dx - \int\left\{\frac{dx}{dx}\int \operatorname{cosec} x . \cot x\, dx\right\} dx$

$= x(-\operatorname{cosec} x) - \int 1 \times (-\operatorname{cosec} x)\, dx$

$= -x\operatorname{cosec} x + \int \operatorname{cosec} x\, dx$

$= -x\operatorname{cosec} x + \ln|\operatorname{cosec} x - \cot x| + c$

e) $\int x \cos^2 x \, dx$

$\cancel{= x \int \cos^2 x \, dx}$

$= \frac{1}{2} \int x \, 2\cos^2 x \, dx$

$= \frac{1}{2} \int x (\cos 2x + 1) \, dx$

$= \frac{1}{2} \int (x + x \cos 2x) \, dx$

$= \frac{1}{2} \int x \, dx + \frac{1}{2} \int x \cos 2x \, dx$

$= \frac{1}{2} \times \frac{x^2}{2} + \frac{1}{2} \left[x \int \cos 2x \, dx - \int \left\{ \frac{dx}{dx} \int \cos 2x \, dx \right\} dx \right]$

$= \frac{1}{4} x^2 + \frac{1}{2} \left[x \frac{\sin 2x}{2} \right] - \frac{1}{2} \int \frac{\sin 2x}{2} \, dx$

$= \frac{x^2}{4} + \frac{1}{2} \times \frac{x}{2} \sin 2x - \frac{1}{4} \left(\frac{-\cos 2x}{2} \right) + c$

$= \frac{x^2}{4} + \frac{x}{4} \sin 2x + \frac{1}{8} \cos 2x + c$

f) $\int x^2 \sin(3x-2) \, dx$

$= x^2 \int \sin(3x-2) - \int \left\{ \frac{dx^2}{dx} \int \sin(3x-2) \, dx \right\} dx$

$= x^2 \frac{(-\cos(3x-2))}{3} - \int \left(2x \frac{(-\cos(3x-2))}{3} \right) dx$

$= -\frac{x^2}{3} \cos(3x-2) + \frac{2}{3} \int x \cos(3x-2) \, dx$

$= -\frac{x^2}{3} \cos(3x-2) + \frac{2}{3} \left[x \int \cos(3x-2) \, dx - \int \left\{ \frac{dx}{dx} \int \cos(3x-2) \, dx \right\} dx \right.$

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$$= \frac{-x^2}{3}\cos(3x-2) + \frac{2}{3}\left[x.\frac{\sin(3x-2)}{3} - \int\frac{\sin(3x-2)}{3}dx\right]$$

$$= \frac{-x^2}{3}\cos(3x-2) + \frac{2x}{9}\sin(3x-2) - \frac{2}{9}\times\left(\frac{-\cos(3x-2)}{3}\right) + c$$

$$= \frac{-x^2}{3}\cos(3x-2) + \frac{2x}{9}\sin(3x-2) + \frac{2}{27}\cos(3x-2) + c$$

(4) Evaluate the following

a) $\int e^x \cos x\, dx = I$ (let)

$$= \int \cos x\; e^x dx$$

$$= \cos x \int e^x dx - \int\left\{\frac{d\cos x}{dx} \times \int e^x dx\right\}dx$$

$$= \cos x \times e^x - \int(-\sin x) \times e^x\, dx$$

$$= \cos x \times e^x + \int \sin x \times e^x\, dx$$

$$= \cos x \times e^x + \left[\sin x \int e^x dx - \int\left\{\frac{d\sin x}{dx} \times \int e^x dx\right\}dx\right]$$

$$= \cos x \times e^x + \left[\sin x \times e^x - \int \cos x \times e^x\, dx\right]$$

$$= \cos x \times e^x + \sin x \times e^x - I$$

Now,

$$I + I = \cos x \times e^x + \sin x \times e^x$$

$$2I = e^x(\cos x + \sin x)$$

$$I = \frac{e^x(\cos x + \sin x)}{2}$$

$$\therefore \int e^x \cos x\, dx = \frac{e^x}{2}(\cos x + \sin x)$$

b) $\int e^{ax} \sin bx \, dx = I$ (let)

$$= \int \sin bx \; e^{ax} \, dx$$

$$= \sin bx \int e^{ax} dx - \int \left\{ \frac{d \sin bx}{dx} \int e^{ax} dx \right\} dx$$

$$= \sin bx \times \frac{e^{ax}}{a} - \int b \cos bx \times \frac{e^{ax}}{a} \, dx$$

$$= \frac{e^{ax}}{a} \sin bx - \frac{b}{a} \int \cos bx \times e^{ax} \, dx$$

$$= \frac{e^{ax}}{a} \sin bx - \frac{b}{a} \left[\cos bx \int e^{ax} dx - \int \left\{ \frac{d \cos bx}{dx} \times \int e^{ax} dx \right\} dx \right.$$

$$= \frac{e^{ax}}{a} \sin bx - \frac{b}{a} \left[\cos bx \times \frac{e^{ax}}{a} - \int -b \sin bx \times \frac{e^{ax}}{a} \, dx \right.$$

$$= \frac{e^{ax}}{a} \sin bx - \frac{b}{a^2} \left[(\cos bx \times e^{ax} + \int b \sin bx \times e^{ax} \, dx \right]$$

$$= \frac{e^{ax}}{a} \sin bx - \frac{b}{a^2} \cos bx \times e^{ax} + \frac{b^2}{a^2} \int \sin bx \times e^{ax} \, dx$$

$$= \frac{e^{ax}}{a} \sin bx - \frac{b}{a^2} \cos bx \times e^{ax} + \frac{b^2}{a^2} \times I + c$$

Now,

$$I = \frac{e^{ax}}{a} \sin bx - \frac{b}{a^2} \cos bx \times e^{ax} + \frac{b^2}{a^2} I + c$$

$$\Rightarrow I - \frac{b^2}{a^2} I = \frac{e^{ax}}{a} \sin bx - \frac{b}{a^2} \cos bx \times e^{ax} + c$$

$$\Rightarrow I \left(\frac{a^2 - b^2}{\cancel{a^2}} \right) = \frac{a e^{ax} \sin bx - b e^{ax} \cos bx + c}{\cancel{a^2}}$$

$$\therefore I = \frac{a e^{ax} \sin bx - b e^{ax} \cos bx + c}{a^2 + b^2}$$

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c) $\int \sec^3 x \, dx = I$ (let)

$= \int \sec^2 x \cdot \sec x \, dx$

$= \int \sec x \cdot \sec^2 x \, dx$

$= \sec x \int \sec^2 x \, dx - \int \left\{ \frac{d \sec x}{dx} \int \sec^2 x \, dx \right\} dx$

$= \sec x \times \tan x - \int \sec x \cdot \tan x \times \tan x \, dx$

$= \sec x \times \tan x - \int \sec x \cdot \tan^2 x \, dx$

$= \sec x \times \tan x - \left[\int \sec x \cdot (\sec^2 x - 1) \cdot dx \right]$

$= \sec x \cdot \tan x - \int (\sec^3 x - \sec x) \, dx$

$= \sec x \cdot \tan x + \int \sec x \, dx - \int \sec^3 x \, dx$

$= \sec x \cdot \tan x + \ln|\sec x + \tan x| - I + c$

Now,

$I = \sec x \cdot \tan x + \ln|\sec x + \tan x| - I + c$

$2I = \sec x \cdot \tan x + \ln|\sec x + \tan x| + c$

$I = \frac{1}{2} (\sec x \cdot \tan x + \ln|\sec x + \tan x| + c$

d) $\int \operatorname{cosec}^3 x \, dx$

let $I = \int \operatorname{cosec}^3 x \, dx$

$= \int \operatorname{cosec} x \cdot \operatorname{cosec}^2 x \, dx$

$= \operatorname{cosec} x \int \operatorname{cosec}^2 x \, dx - \int \left\{ \frac{d \operatorname{cosec} x}{dx} \cdot \int \operatorname{cosec}^2 x \, dx \right\} dx$

$= \operatorname{cosec} x \cdot (-\cot x) - \int \{ -\operatorname{cosec} x \cdot \cot x \cdot (-\cot x) \} dx$

$= -\operatorname{cosec} x \cdot \cot x - \int \{ \operatorname{cosec} x \cdot \cot^2 x \} dx$

$= -\operatorname{cosec} x \cdot \cot x - \int \operatorname{cosec} x (\operatorname{cosec}^2 x - 1) \, dx$

$= -\operatorname{cosec} x \cdot \cot x - \int \operatorname{cosec}^3 x \, dx + \int \operatorname{cosec} x \, dx$

$$I = -\operatorname{cosec}x \cdot \cot x - I + \ln|\operatorname{cosec}x - \cot x| + c$$

$$I + I = -\operatorname{cosec}x \cdot \cot x + \ln|\operatorname{cosec}x - \cot x| + c$$

$$I = -\frac{1}{2}\left[\operatorname{cosec}x \cdot \cot x - \ln|\operatorname{cosec}x - \cot x|\right] + c \quad 4$$

⑤ Evaluate the following definite integral

a) $\int_1^e x \ln x \, dx$

$$\int x \ln x \, dx$$

$$= \int \ln x \cdot x \, dx$$

$$= \ln x \int x \, dx - \int\left\{\frac{d \ln x}{dx} \cdot \int x \, dx\right\} dx$$

$$= \ln x \times \frac{x^2}{2} - \int \frac{1}{\cancel{x}} \times \frac{x^{\cancel{2}}}{2} \, dx$$

$$= \ln x \times \frac{x^2}{2} - \frac{1}{2}\int x \, dx$$

$$= \ln x \times \frac{x^2}{2} - \frac{1}{2} \times \frac{x^2}{2}$$

$$= \frac{\ln x \cancel{\times} x^2}{2} - \frac{1}{4}x^2$$

Now,

$$\int_1^e x \ln x \, dx$$

$$= \cancel{\int_1^e \frac{\ln x\, x^2}{2} - \frac{1}{4}}\left[\ln x \cdot \times \frac{x^2}{2} - \frac{1}{4}x^2\right]_1^e$$

$$= \ln e \times \frac{e^2}{2} - \frac{1}{4}e^2 - \ln 1 \times \frac{1}{2} + \frac{1}{4} \times 1^2$$

$$= 1 \times \frac{e^2}{2} - \frac{1}{4}e^2 - 0 + \frac{1}{4}$$

$= \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4}$

$= \frac{e^2}{4} + \frac{1}{4}$

$= \frac{1}{4}(e^2+1)$ //

b) $\int_0^1 x e^x dx$

$\int x e^x dx$

$= x \int e^x dx - \int \left\{ \frac{dx}{dx} \times \int e^x dx \right\} dx$

$= x \times e^x - \int 1 \times e^x \, dx$

$= x \times e^x - e^x$

$= e^x (x-1)$

Now,

$\int_0^1 x e^x dx$

$= \left[e^x (x-1) \right]_0^1$

$= e^1 (1-1) - e^0 (0-1)$

$= e \times 0 - 1 \times -1$

$= 0 + 1$ //

c) $\int_0^{\pi/4} x \sec^2 x \, dx$

$\int x \sec^2 x \, dx$

$= x \int \sec^2 x \, dx - \int \left\{ \frac{dx}{dx} \int \sec^2 x \, dx \right\} dx$

$= x \tan x - \int 1 \times \tan x \, dx$

$= x\tan x - \ln|\sec x| \cancel{+}$

$\int_0^{\pi/4} x\sec^2 x\,dx$

$= \left[x\tan x - \ln|\sec x|\right]_0^{\pi/4}$

$= \frac{\pi}{4}\tan\frac{\pi}{4} - \ln\left|\sec\frac{\pi}{4}\right| - 0.\tan 0 + \ln|\sec 0|$

$= \frac{\pi}{4} - \ln\sqrt{2} - 0 + \ln 1$

$= \frac{\pi}{4} - \frac{1}{2}\ln 2 + 0 \quad [\because \ln 1 = 0]$

$= \frac{\pi}{4} - \frac{1}{2}\ln 2$

d) $\int_0^{\pi} x\sin^2 x\,dx$

$\int x\sin^2 x\,dx$

$= \frac{1}{2}\int x\, 2\sin^2 x\,dx$

$= \frac{1}{2}\int x(1-\cos 2x)\,dx$

$= \frac{1}{2}\int (x - x\cos 2x)\,dx$

$= \frac{1}{2}\int x\,dx - \frac{1}{2}\int x\cos 2x\,dx$

$= \frac{1}{2}\times\frac{x^2}{2} - \frac{1}{2}\left[x\int\cos 2x\,dx - \int\left\{\frac{dx}{dx}\times\int\cos 2x\,dx\right.\right.$

$= \frac{1}{4}x^2 - \frac{1}{2}\left[x\frac{\sin 2x}{2} - \int 1\times\frac{\sin 2x}{2}\,dx\right.$

$= \frac{1}{4}x^2 - \frac{1}{4}x\sin 2x - \frac{1}{4}\frac{\cos 2x}{2}$

$= \frac{1}{4} x^2 - \frac{1}{4} \sin 2x - \frac{1}{8} \cos 2x$

$= \frac{1}{4} \left[x^2 - \sin 2x - \frac{1}{2} \cos 2x \right]$

Now,

$\int_0^{\pi} x \sin^2 x$

$= \frac{1}{4} \left[x^2 - \sin 2x - \frac{1}{2} \cos 2x \right]_0^{\pi}$

$= \frac{1}{4} \left[\pi^2 - \sin 2\pi - \frac{1}{2} \cos 2\pi - 0 + \sin 0 + \frac{1}{2} \cos 0 \right]$

$= \frac{1}{4} \left[\pi^2 - 0 - \cancel{\frac{1}{2}} + 0 + 0 + \cancel{\frac{1}{2}} \right]$

$= \frac{1}{4} (\pi^2)$

$= \frac{\pi^2}{4}$ Ans.